



NATIONAL UNIVERSITY OF TECHNOLOGY

ICAT PROJECT REPORT

ICAT PROJECT TITLE:

CHESS GAME WITH AI

ICAT PROJECT MEMBERS:

- HARIS HAMID
- MUHAMMAD MURSAL
- FARHAN AWAN
- HAFSA HASNAIN
- SAMEEN ZAHID

ICAT PROJECT SUPERVISOR:

- Lec. Hajra Ahmad

DESCRIPTION:

This project is a chess game also Pygame has been used in its development and it offers two game modes: PvP and PvAI. The time control can be chosen and the game starts from the interface which has a start menu where players can select the mode. There are two time control settings available for the game: 10 minutes and 30 minutes. It has the capability to play either against a fellow player or the computer using an AI opponent from Stockfish.

It is standard chessboard with all the special moves like castling, en passant, and pawn promotion. This makes the AI to evaluate the position on the board and respond with the best move possible according to the Stockfish engine. It has a very simple and clean interface with a clear board and buttons to set up the game to make it easy to use. This project shows a full development of the game, AI, and UI, thus it is a good example of the digital chess implementation.

PROJECT CONTRIBUTION:

M.MURSAL:

- CORE FUNCTIONALITY
- PIECE LEGAL MOVEMENT
- PAWN PROMOTION
- STOCKFISH AI INTEGRATION
- KING IN CHECK
- PIECES CAUGHT TAB
- UI INTEGRATION
- GAME DEBUGGING AND ERROR FIXING

FARHAN AWAN:

- PIECE MOVES SUGGESTION
- LIST OF ALL THE PIECES
- GAME TESTING AND ERROR FINDING

HARIS HAMID:

- GAME TIMER
- CHESS BOARD DRAW IN PYGAME
- GAME PROTOTYPE

HAFSHA:

- CHECK BOX

SAMEEN:

- BUTTON FOR MENU